

BM4 Implicit Method

Physical state and one reduced Hairer projection per complete cycle

1 Twelve-stage BM4 base map

Let χ_h be a first-order map and $\chi_h^* = \chi_h^{-1}$ its adjoint. One complete step uses the palindromic composition

$$\Psi_h = \chi_{b_{12}h} \circ \chi_{b_{11}h}^* \circ \cdots \circ \chi_{b_2h} \circ \chi_{b_1h}^*,$$

with alternating adjoint and direct maps in chronological order and

$$(b_1, \dots, b_{12}) = (a_1, a_2, a_3, a_4, a_5, a_6, a_6, a_5, a_4, a_3, a_2, a_1),$$

$$\begin{aligned} a_1 &= 0.0792036964311957, & a_2 &= 0.1303114101821663, \\ a_3 &= 0.2228614958676077, & a_4 &= -0.3667132690474257, \\ a_5 &= 0.3246481886897062, & a_6 &= 0.1096884778767498. \end{aligned}$$

The half-cycle coefficients sum to $1/2$. Palindromy makes the composition self-adjoint and the order conditions give designed global order four. The negative coefficient a_4 is an intentional backward subflow.

2 Physical state and internal workspace

Twelve direct/adjoint stages enclosed by one spatial Hairer projection. Energy tracking is optional and passive. After convergence, passive energy quadrature evaluates $-\partial_t H$ at the 24 shear inputs retained from the accepted residual, with their original signed durations. It repeats no spatial map or coupling and performs no new nonlinear solve. Trial traces are discarded; energy evaluations remain outside physical solver counters. The common state conventions are specified below.

3 One Hairer projection around the complete cycle

For every complete BM4 step, the method displaces the diagonal embedding by a single multiplier $\mu \in \mathbb{R}^{2N}$,

$$\hat{Y}_n = Ez_n + N\mu, \quad M(\mu) = \Psi_{h,t_n}(\hat{Y}_n), \quad (1)$$

$$r(\mu) = GM(\mu) + 2\mu, \quad z_{n+1} = P(M(\mu) + N\mu). \quad (2)$$

The accepted multiplier is the root $r(\mu) = 0$. Thus the projection is applied once around the full twelve-stage composition, never between its internal stages.

With $J_{\text{BM4}} = D\Psi_{h,t_n}(\hat{Y}_n)$, Newton uses

$$D_\mu r = GJ_{\text{BM4}}N + 2I.$$

The implementation forms J_{BM4} as the exact ordered product of the twelve guiding-centre stage Jacobians, with a centred-difference fallback. Good Broyden is retained as a nonlinear-solver choice; it changes the finite iteration path but not the defining projected map.

4 Fixed-grid lifecycle and diagnostics

The fixed controller advances accepted diagonal states on an output-independent grid. Off-grid samples use independent shadow steps. Only main steps contribute observations and nonlinear diagnostics: corrections, residual evaluations, residual norm, tolerance and multiplier norm. Coupling and Jacobian settings are recorded as metadata. Optional observers retain physical endpoints and twelve base-stage snapshots, allowing an independent energy reconstruction.

5 Implementation correspondence

The method is in `src/simulation/methods/bm4/implicit.py`; `_core.py` owns the coefficient schedule and twelve-stage traversal. See `implicit-reduced.tex` for the reduced-projection derivation and `../simulation/` for runtime contracts.

6 Four state formulations and passive energy tracking

For one planar particle the supported internal representations are

physical	$z = (x, y)$	$\mathbb{R}^2$,
physical with energy	(z, t, κ)	$\mathbb{R}^4$,
duplicated	$(u, v), \quad u, v \in \mathbb{R}^2$	$\mathbb{R}^4$,
duplicated with energy	(u, v, t, κ)	$\mathbb{R}^6$.

Classical methods use the physical rows; ABBA and BM4 use the duplicated rows. `track_energy` selects the energy variant. For N planar particles the respective dimensions are $2N$, $4N$, $4N$, and $6N$: one time entry and one momentum per particle. All time entries equal the integration time. Classical full-cyclotron runs instead use $4N$ or $6N$ entries. Accepted spatial copies coincide; only one physical copy is exported. Every coordinate block contains N values in component-major order. The formulation validates particle times and aligns output round-off; the integrator does not interpret coordinate indices.

The normalized passive balance is

$$\dot{\kappa} = -\partial_t H(t, z), \quad \kappa(t_0) = 0, \quad \Delta K(t) = H(t, z(t)) + \kappa(t) - H(t_0, z_0).$$

Splitting accumulators from both copies are divided by two. The physical Hamiltonian may vary with time; ΔK measures its computed balance. Tracking leaves physical trajectories, nonlinear stopping rules and adaptive acceptance unchanged. Dynamics must supply the momentum derivative, including zero for an autonomous Hamiltonian.

Hairer's constraint is solely $u - v = 0$: the reduced multiplier has $2N$ entries and ABBA's simultaneous spatial solve has $6N$ unknowns. Neither t nor κ is projected. The former eight-component projection is retired; `state_extension='fully_extended'` raises migration guidance.

Dynamics, formulation, method and integration remain separate. The shared state classes own layouts, diagonal embedding and history extraction in

`src/simulation/formulations/state.py`

Each integration initializes a fresh method instance. Observers receive physical states; energy histories are in `Solution.diagnostics`. The `extended_time` history has shape (N, n_{samples}) . Fixed-step shadow samples leave main states and counters unchanged. Adaptive energy quadrature uses accepted dense output; its backend state and Jacobian remain physical.